

Plane strain stress intensity factors for branched cracks

V. VITEK

Materials Division, Central Electricity Research Laboratories, Leatherhead, Surrey, KT 22 7SE, UK

(Received November 18, 1976; in revised form February 18, 1977)

ABSTRACT

A method of calculating stress intensity factors for branched and bent cracks embedded in an infinite body has been developed. The branches are always assumed to be sharp cracks and are modelled by dislocation distributions. The original crack may be either sharp or of elliptical cross-section with finite root radius. Hence, the method which has a precision $\pm 2\%$, is also applicable to the study of crack branches emanating from elliptical holes and, approximately, also from notches. The following detailed calculations have been made assuming mode I loading: branched sharp crack with branches of equal and different length, bent sharp crack, and one and two crack branches emanating from the crack with a finite root radius. Bending of a sharp crack under mixed mode loading has also been studied. The criteria of maximum tensile stress and maximum energy release rate used in the study of direction of crack propagation are discussed.

1. Introduction

Branching of propagating cracks is a familiar phenomenon in fast brittle fracture (e.g. [1]) and in stress corrosion cracking [2–5]. The deflection of a crack out of its original plane to form a bent crack may be regarded as a special case of branching, and usually takes place under combined mode I and mode II loading, for example when a crack is loaded in tension with the tensile axis not perpendicular to the crack plane [6–8]. In connection with the study of these phenomena calculations of stress intensity factors and energy release rates have been made by a number of authors for the case of antiplane strain [9–12]. However, for the case of plane strain, which is usually encountered in practical situations, only very limited studies have been made so far. Andersson [13] attempted to solve this problem using an irrational mapping function which maps a general star-shaped crack onto a circle [14], and the theory of complex potentials [15]. However, after correcting a vital mistake [16] it was concluded that the calculation is only applicable for infinitesimally small branches. A similar method was used in [8] in the study of bent cracks, assuming that the deviating segment was always much smaller than the original crack. A limited study of cracks with long branches was carried out in [4, 17, 18]. The same mapping function as in [13] was employed and expanded into an infinite series which is then truncated in the numerical calculation.

In this paper a general method employing a dislocation model is developed for calculating stress intensity factors of branched cracks embedded in an infinite body. It is applicable to any length of branches and, furthermore, the original crack can be of elliptical cross-section with finite root radius. Thus the method can also be used to calculate the stress intensity factors of crack branches emanating from elliptical holes. A number of special cases of branched and bent cracks is studied in detail.

2. Theory

The configuration of a crack with branches is shown in Fig. 1. The original crack is of elliptical cross-section with major semi-axis a in the x direction and minor semi-axis b

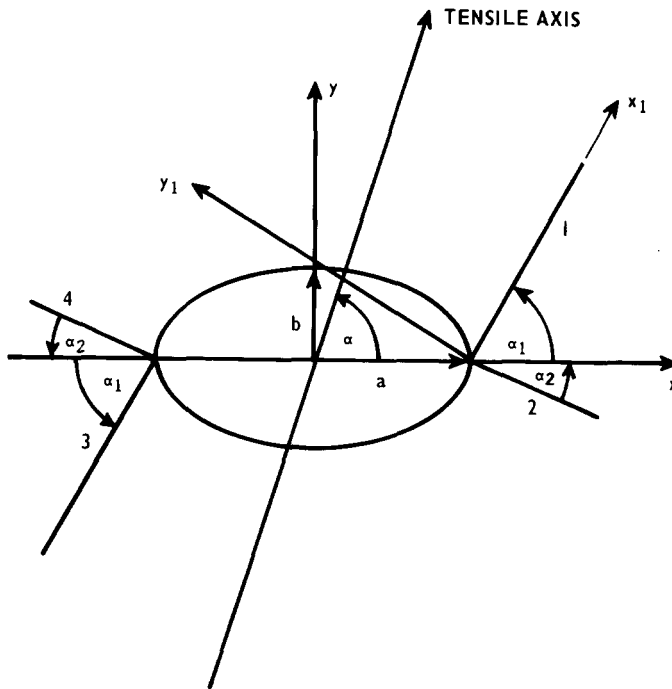


Figure 1. Schematic picture of the crack of elliptical cross-section with crack branches emanating from the tips.

in the y direction. The branches are sharp cracks of lengths a_1 and a_2 inclined to the plane of the original crack by angles α_1 and α_2 , respectively. All dimensions are infinite in the z direction and plane strain deformation is assumed.* The crack is embedded in an infinite elastic isotropic body which is subject to an applied uniaxial tensile stress σ with the tensile axis inclined by the angle α to the x axis. The configuration of the branching crack is always assumed to be symmetrical with respect to the plane $x=0$. This means that in the case of an embedded crack branching is supposed to take place at both crack tips. At the same time this also provides an approximation for a crack emanating from a free surface ($x=0$). The part of the branching crack in the half space $x < 0$ is then regarded as an image which to a good approximation cancels tractions on the free surface. In the following the branches will be numbered as shown in Fig. 1. Each branch will be represented by distributions of edge dislocations of two types. One type of dislocations D_j^1 ($j = 1, \dots, 4$), have Burgers vectors perpendicular to the branch, i.e. in the direction of the axis y_1 for the branch 1 and the other, D_j^2 parallel to the branch, i.e. in the direction of the axis x_1 for branch 1. In the coordinate system x_1y_1 (Fig. 1) the dislocation distribution near the tip of the branch 1 is [19].

$$D_1^1(x_1) = \frac{2(1-\nu)}{G} \frac{K_I^1}{\sqrt{2\pi(a_1-x_1)}} \quad (1)$$

where G is the shear modulus and ν is the Poisson ratio. K_I^1 and K_{II}^1 are corresponding stress intensity factors for the branch 1. The same applies to other branches. Thus, if the dislocation distributions are known the stress intensity factors can be calculated using Eqn. (1).

* Note that z is also used later to denote the complex variable $x + iy$.

The dislocation distributions in the branches must satisfy the conditions of zero tractions on each branch. For branch k in the coordinates x_k, y_k (these refer to local reference axes similar to x_1, y_1 for branch 1) these give rise to the following equations:

$$\tau_{22}^{c+appl}(x_k) + \sum_{j=1}^4 \sum_{i=1}^2 \int_0^{a_j} D_j^i(x_j) [\tau_{22}^{d,i}(x_k, x_j) + \tau_{22}^{dim,i}(x_k, x_j)] dx_j = 0 \quad (2a)$$

$$\tau_{12}^{c+appl}(x_k) + \sum_{j=1}^4 \sum_{i=1}^2 \int_0^{a_j} D_j^i(x_j) [\tau_{12}^{d,i}(x_k, x_j) + \tau_{12}^{dim,i}(x_k, x_j)] dx_j = 0 \quad (2b)$$

Due to the symmetry $D_3^i = -D_1^i$ and $D_4^i = -D_2^i$, $\tau_{mn}^{c+appl}(x_k)$ is the stress tensor due to the crack and applied stress. $\tau_{mn}^{d,i}(x_k, x_j)$ is the stress tensor due to a dislocation of unit Burgers vector of appropriate direction, situated at x_j in the j th branch; $\tau_{mn}^{dim,i}(x_k, x_j)$ is the image stress tensor of this unit dislocation arising from the condition of zero tractions on the surface of the original crack. Details of the calculation of these stresses are given in Appendix I.

Equations (2a, b) represent a system of singular, integral equations which cannot be solved analytically. A numerical solution can be found by replacing the integral equations by an appropriate system of linear equations. To do this we proceed as follows:

First we divide each branch j into n intervals of length Δx_j and assume that the dislocation content in an interval l is $b_{j,l}^i$ ($i = 1, 2$). The Eqns. (2ab) can then be written as a system of $2n$ linear equations

$$\tau_{22}^{c+appl}(x_k^r) + \sum_{l=1}^n \sum_{j=1}^2 \sum_{i=1}^2 b_{j,l}^i [\tau_{22}^{d,i}(x_k^r, x_j^l) - \tau_{22}^{d,i}(x_k^r, x_{j+2}^l) + \tau_{22}^{dim,i}(x_k^r, x_j^l) - \tau_{22}^{dim,i}(x_k^r, x_{j+2}^l)] = 0 \quad (3a)$$

$$\tau_{12}^{c+appl}(x_k^r) + \sum_{l=1}^n \sum_{j=1}^2 \sum_{i=1}^2 b_{j,l}^i [\tau_{12}^{d,i}(x_k^r, x_j^l) - \tau_{12}^{d,i}(x_k^r, x_{j+2}^l) + \tau_{12}^{dim,i}(x_k^r, x_j^l) - \tau_{12}^{dim,i}(x_k^r, x_{j+2}^l)] = 0 \quad (3b)$$

where $r = 1, \dots, n$ and $k = 1, 2$. x_k^r means the coordinate of the centre of interval s on branch t ; $\tau_{mn}^{d,i}(x_k^r, x_j^l)$ and $\tau_{mn}^{dim,i}(x_k^r, x_j^l)$ are, as in Eqns. (2a, b) the stress tensor and the image stress tensor, respectively, at the point x_k^r due to the unit dislocation distributed in the interval l on the branch j . Details of the calculation of these stress tensors and of the numerical solution of Eqn. (3) are given in Appendix II.

Numerical solution of Eqns. (3) gives the dislocation contents $b_{j,l}^i$ and the stress intensity factors are determined by the dislocation contents in the intervals at the tip of the branches, i.e. intervals n . Using (1) we obtain

$$K_i^j = \frac{G\sqrt{2\pi}}{4(1-\nu)\sqrt{\Delta x_j}} b_{j,n}^i \quad (4)$$

The method was first tested for the case of a sharp original crack ($b = 0$) with only branches 1 and 3 and tensile axis perpendicular to the crack ($\alpha = 90^\circ$). For $\alpha_1 = 0$ this represents an extension of the original crack and $K_I = \sigma\sqrt{\pi(a+a_1)}$ was found within 2%. For a case of $a_1 \gg a$ and $\alpha_1 \neq 0$ configuration is virtually equivalent to a crack of length $2a_1$ inclined to the tensile axis by an angle α_1 . For $a_1/a > 50$ the numerical results agreed within 2% with the analytical solutions $K_I = \sigma\sqrt{\pi a_1} \cos^2 \alpha_1$ and $K_{II} = \sigma\sqrt{\pi a_1} \cos \alpha_1 \sin \alpha_1$. Comparisons with previous calculations have also been made and they are presented in the following section.

3. Results

The stress intensity factors were calculated numerically for six cases of branching and bending cracks which are shown schematically in Fig. 2; the value $\nu = 1/3$ was taken in all these calculations. In the first five cases (Figs. 2a–e) the tensile axis is perpendicular to the plane of the original crack ($\alpha = 90^\circ$) i.e. mode I loading is assumed. In the last case (Fig. 2f) the tensile axis is inclined by an angle $\alpha < 90^\circ$ to the plane of the original crack, i.e. a mixed mode loading. In the following the results for each of these cases are reviewed.

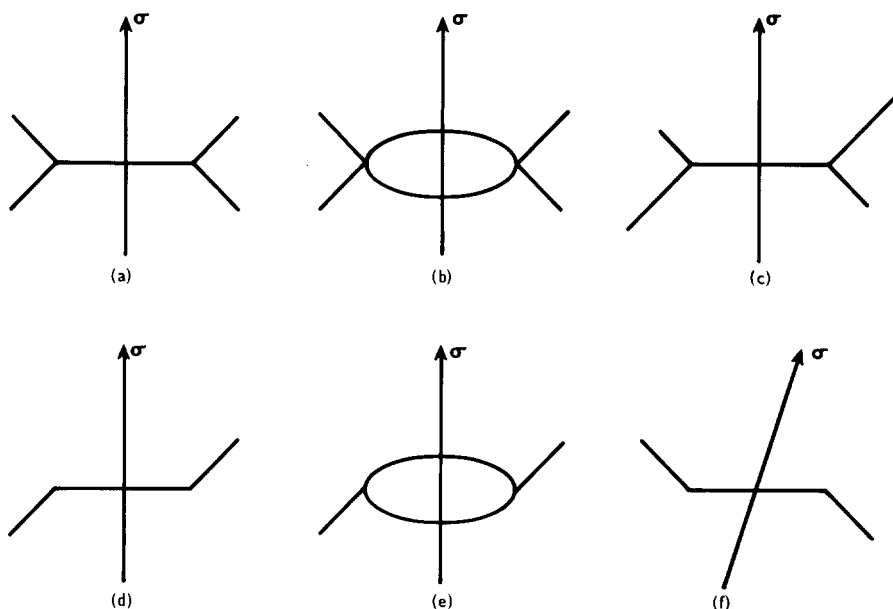


Figure 2. Schematic representation of the six cases of branched and bent cracks studied in this paper.

3.1. Symmetrically branching sharp crack (Fig. 2a)

The branches are symmetrical, i.e. $a_1 = a_2$ and $\alpha_1 = \alpha_2$. Obviously, the stress intensity factors are the same for both branches and we shall, therefore, omit the branch index. Their dependence on the length of the branches is shown in Figs. 3a, b for different angles α_1 and their dependence on the inclination of the branches, α_1 , for different branch lengths is shown in Figs. 3c, d (full lines). Since this type of crack branching is often encountered in practice (fast fracture, stress corrosion cracking) all the calculated values of K_I and K_{II} are tabulated in Tables 1a, b; linear interpolation for other angles and lengths of the branches can be made.

This type of crack branching was also studied theoretically in [4] but in this calculation branches were present only on one side of the sharp crack. Nevertheless, the results of [4] can be compared with ours for branches of small length since in this case the interaction of branches 1 and 2 with branches 3 and 4 is small. The dependences of K_I and K_{II} on α_1 are for $a_1/a = 0.2$ and $\alpha_1 > 10^\circ$ shown in Fig. 14 of Ref. [4] and when this is compared with Figs. 3a, b the results are found to be the same within 5%.

For values of $\alpha_1 \rightarrow 0$, K_I and K_{II} converge to the limits $\frac{1}{2}\sigma\sqrt{\pi(a+a_1)}$ and 0

respectively which correspond to extending the crack by a_1 . However, for small angles α_1 some of the dislocations in the two branches come so close to each other that significant numerical difficulties are encountered in evaluating their interaction. Hence, the calculation was not carried out for very small α_1 and, therefore, the dependence of K_I and K_{II} on α_1 for $0 < \alpha_1 < 5^\circ$ was not obtained. It is seen from Fig. 3a that for $\alpha_1 > 30^\circ$, K_I first decreases with the branch length and starts to increase again only when the length of the branches is of the same order of magnitude as the length of the original crack. K_{II} of course always increases with the length of the branches. $(K_I)^2 + (K_{II})^2$ which measures the energy release rate, G , in the direction of the branch ($G = (1 - \nu^2/E)(K_I)^2 + (K_{II})^2$, where E is the Young modulus) is substantially smaller for a branched crack than for the original crack when $a_1 < a$ and always smaller than for the equivalent straight crack of length $a + a_1 \cos \alpha_1$. Hence, whether it is the maximum tensile stress [6] or G [8] which governs the propagation of cracks, this kind of branching will always retard the propagation.

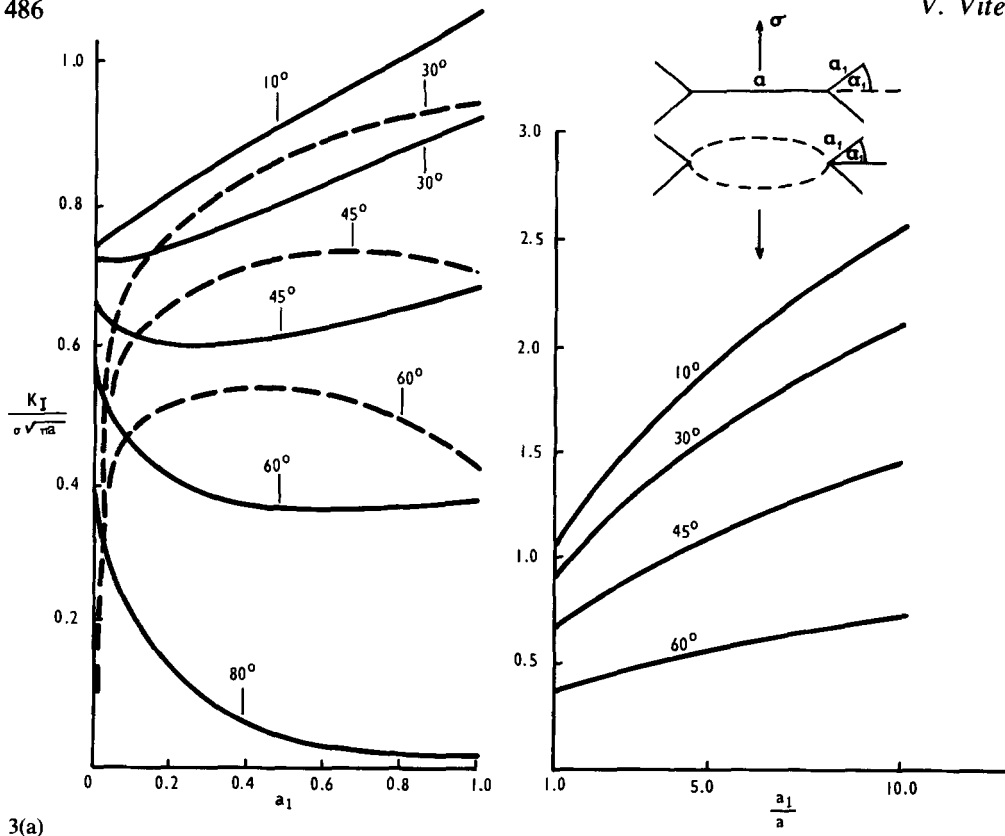
It is seen from Figs. 3c, d that K_{II} changes sign and K_I reaches a maximum for $8^\circ < \alpha_1 < 24^\circ$ depending on the branch length.* The absolute maximum value of K_I is, of course, for $\alpha_1 = 0$ but as mentioned above, the calculations were not performed for $\alpha_1 < 5^\circ$ and thus the maximum found is a local one. This agrees with finite element computations of Kalthoff [20] who studied crack branches emanating from a free surface and found that K_{II} changes sign for $\alpha_1 = 14^\circ$. Crack branching in this case cannot, therefore, be explained on the basis of continuum mechanics only and the reasons for its existence must be metallurgical. This may be, for example, preferable anodic dissolution along the branches of the plastic zone during stress corrosion cracking. We may assume, however, that after branching has occurred the branches will tend to extend in the direction perpendicular to the maximum tensile stress, i.e. where $K_{II} = 0$. Then α_1 will tend to increase if $K_{II} < 0$ and decrease if $K_{II} > 0$. Hence, the branching angle $\alpha_1 \approx 20^\circ$ will be the most favourable for small branches and it will decrease to $\alpha_1 \approx 10^\circ$ when the length of the branches is comparable with the length of the original crack.

3.2. Symmetrical crack branches emanating from an elliptical crack (Fig. 2b)

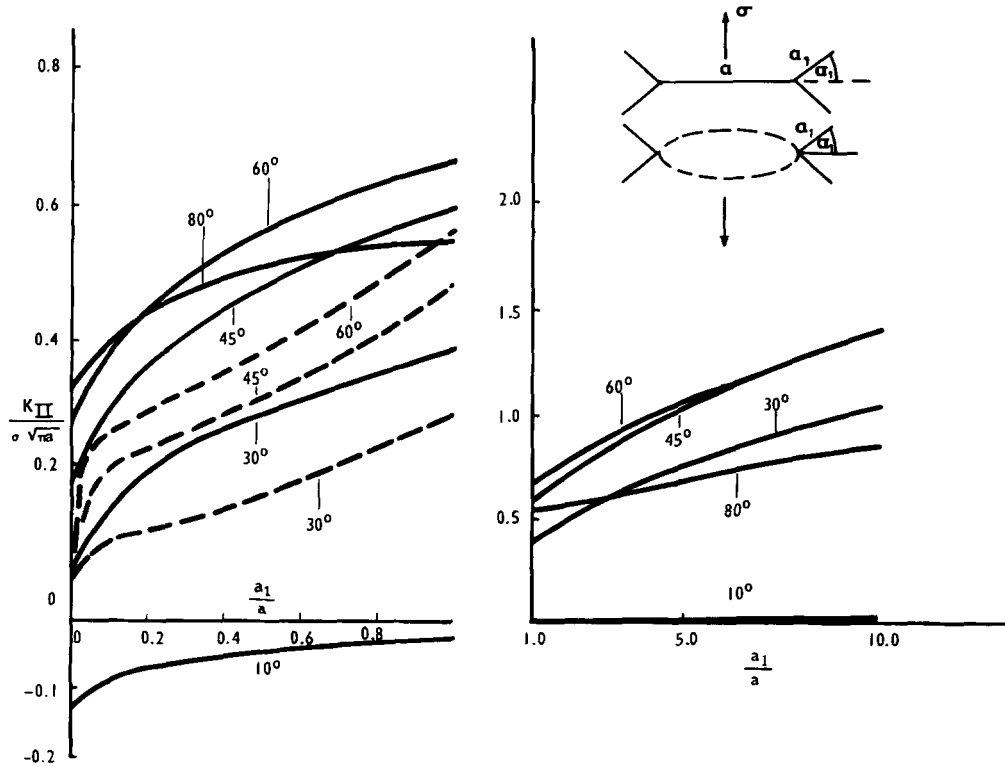
As in the previous case $a_1 = a_2$ and $\alpha_1 = \alpha_2$ and the stress intensity factors are therefore again the same for both branches. The calculation was carried out for a number of inclination angles α_1 and lengths of the branches a_1 and also for different root radii $\rho = b^2/a$ of the elliptical crack. Typical results are again shown in Figs. 3a–d for $\rho/a = 0.5$ (dashed lines).

It is seen that for small a_1 both K_I and K_{II} are much smaller than in the case of the sharp crack and converge to zero as $a_1 \rightarrow 0$. However, when the length of the branches increases K_I increases above the value corresponding to the sharp original crack and approaches this value asymptotically for $a_1 \gg \rho$. K_{II} is always smaller than in the case of the sharp original crack and when the branches are long it again asymptotically approaches this value. This behaviour results from the fact that at distances comparable with the crack length the stress is elevated more strongly ahead of a blunt crack of elliptical cross-section than ahead of a sharp crack of the same

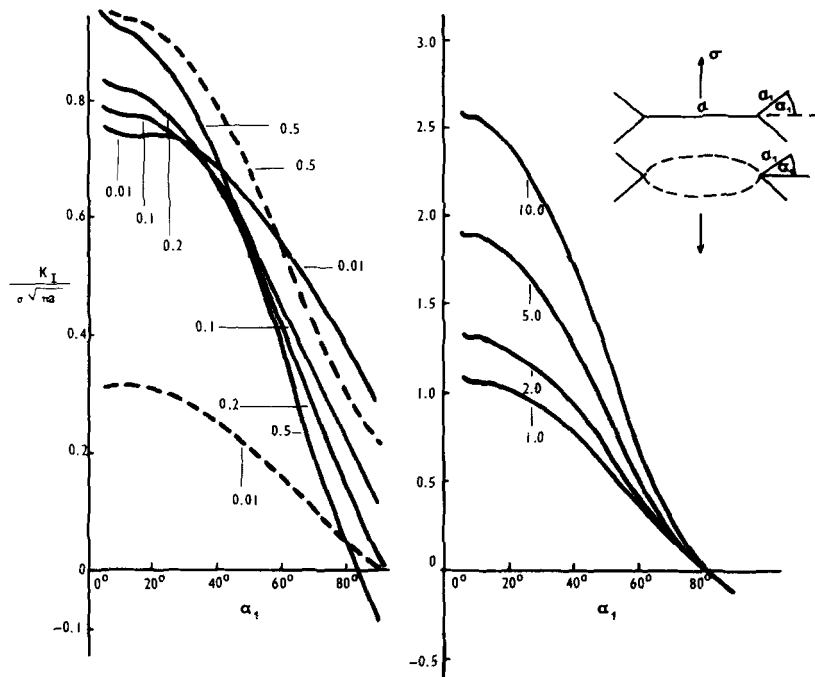
* K_I also changes sign and becomes negative for the case of long branches and $\alpha_1 > 80^\circ$. However, negative K_I is not physically meaningful and in fact the crack is closed when $K_I = 0$. The reason for the negative values is purely mathematical since zero tractions are imposed on the original crack surface even when it is closed although physically these are no more required.



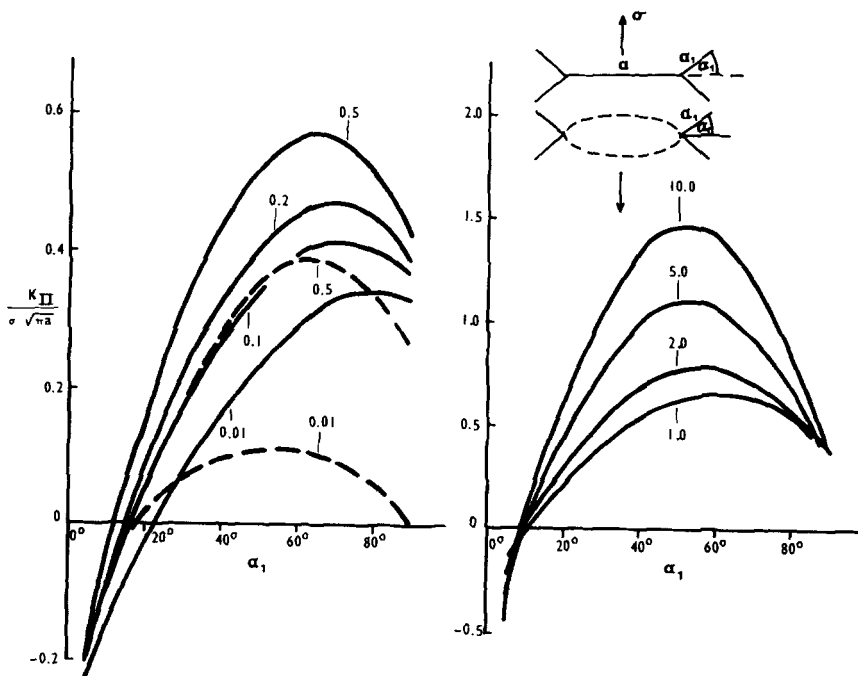
3(a)



3(b)



3(c)



3(d)

Figure 3. Stress intensity factors for symmetrically branched sharp crack, Fig. 2a (full lines); and symmetrical crack branches emanating from an elliptical crack, Fig. 2b (dashed lines); (a) K_I vs. a_1/a for different angles α_1 ; (b) K_{II} vs. a_1/a for different angles α_1 ; (c) K_I vs. the angle α_1 for different values of a_1/a ; (d) K_{II} vs. the angle α_1 for different values of a_1/a .

TABLE 1
Stress intensity factors for symmetrically branched crack; (a) $K'_I = K_{II}/\sigma\sqrt{\pi a}$; (b) $K'_{II} = K_{III}/\sigma\sqrt{\pi a}$.

TABLE 1(a)

α_1 a_1/a	5°	7.5°	10°	12.5°	15°	20°	25°	30°	40°	45°	50°	60°	70°	80°	90°
0.1	0.752	0.743	0.740	0.738	0.735	0.738	0.736	0.724	0.688	0.662	0.631	0.557	0.471	0.380	0.289
0.05	0.771	0.761	0.756	0.753	0.752	0.746	0.735	0.718	0.663	0.627	0.586	0.491	0.387	0.280	0.180
0.10	0.792	0.781	0.776	0.772	0.773	0.761	0.745	0.723	0.654	0.611	0.562	0.452	0.333	0.216	0.108
0.2	0.831	0.820	0.814	0.809	0.812	0.791	0.770	0.740	0.654	0.600	0.541	0.410	0.272	0.139	0.021
0.3	0.869	0.857	0.850	0.844	0.842	0.822	0.797	0.761	0.662	0.601	0.535	0.388	0.237	0.093	-0.031
0.5	0.938	0.925	0.917	0.918	0.903	0.882	0.850	0.807	0.688	0.616	0.538	0.370	0.202	0.046	-0.084
0.7	1.002	0.989	0.980	0.983	0.963	0.939	0.903	0.853	0.719	0.638	0.552	0.369	0.188	0.027	-0.102
1.0	1.091	1.077	1.067	1.068	1.048	1.019	0.977	0.919	0.767	0.676	0.579	0.377	0.184	0.017	-0.108
1.5	1.225	1.210	1.198	1.195	1.175	1.141	1.090	1.022	0.844	0.739	0.628	0.401	0.188	0.013	-0.110
2.0	1.346	1.318	1.316	1.304	1.290	1.251	1.193	1.116	0.916	0.799	0.675	0.425	0.195	0.010	-0.111
3.0	1.559	1.535	1.533	1.510	1.493	1.446	1.377	1.285	1.047	0.908	0.764	0.472	0.209	0.006	-0.113
5.0	1.910	1.894	1.890	1.855	1.834	1.774	1.685	1.569	1.269	1.096	0.915	0.555	0.237	0.002	-0.115
7.0	2.213	2.180	2.177	2.145	2.120	2.050	1.946	1.809	1.458	1.256	1.046	0.664	0.263	-0.001	-0.118
10.0	2.598	2.564	2.558	2.518	2.489	2.405	2.282	2.120	1.703	1.464	1.216	0.725	0.298	-0.004	-0.122

TABLE I(b)

a_1/a	5°	7.5°	10°	12.5°	15°	20°	25°	30°	40°	45°	50°	60°	70°	80°	90°
0.01	-0.219	-0.163	-0.124	-0.094	-0.069	-0.021	0.026	0.073	0.161	0.200	0.235	0.292	0.327	0.339	0.327
0.05	-0.209	-0.146	-0.102	-0.068	-0.038	0.0181	0.072	0.125	0.221	0.262	0.298	0.350	0.376	0.374	0.346
0.10	-0.203	-0.135	-0.088	-0.050	-0.017	0.045	0.105	0.162	0.264	0.307	0.343	0.394	0.414	0.402	0.362
0.2	-0.199	-0.124	-0.071	-0.028	0.009	0.080	0.147	0.211	0.323	0.368	0.405	0.454	0.466	0.442	0.387
0.3	-0.199	-0.118	-0.060	-0.013	0.0280	0.105	0.178	0.247	0.366	0.413	0.451	0.498	0.504	0.471	0.405
0.5	-0.202	-0.111	-0.046	0.007	0.054	0.142	0.224	0.301	0.431	0.481	0.520	0.562	0.558	0.510	0.427
0.7	-0.207	-0.108	-0.036	0.022	0.073	0.169	0.259	0.342	0.481	0.532	0.572	0.609	0.594	0.532	0.434
1.0	-0.217	-0.106	-0.026	0.038	0.096	0.203	0.302	0.393	0.541	0.594	0.633	0.663	0.633	0.551	0.431
1.5	-0.233	-0.107	-0.016	0.058	0.124	0.246	0.359	0.460	0.621	0.677	0.714	0.733	0.680	0.570	0.419
2.0	-0.250	-0.110	-0.008	0.074	0.147	0.282	0.406	0.516	0.689	0.746	0.783	0.792	0.722	0.586	0.410
3.0	-0.281	-0.117	0.002	0.098	0.184	0.341	0.484	0.611	0.804	0.865	0.901	0.896	0.797	0.622	0.400
5.0	-0.336	-0.132	0.015	0.134	0.240	0.433	0.608	0.762	0.912	1.061	1.099	1.075	0.932	0.692	0.389
7.0	-0.383	-0.147	0.024	0.162	0.285	0.509	0.711	0.888	1.149	1.226	1.265	1.228	1.050	0.757	0.383
10.0	-0.445	-0.167	0.034	0.196	0.341	0.604	0.841	1.048	1.351	1.438	1.480	1.427	1.205	0.845	0.376

length. As for the sharp crack K_{II} changes sign and K_I reaches a maximum for similar values of α_1 and hence the most favourable branching angles are also similar. Results of this calculation also apply approximately to the symmetrical crack branches emanating from a notch of elliptical cross-section with the root radius ρ .

3.3. Asymmetrically branched sharp crack (Fig. 2c)

The inclination of the two branches is the same, i.e. $\alpha_1 = \alpha_2$ but the branch 1 is assumed to be longer than branch 2, i.e. $a_1 > a_2$. The stress intensity factors are now

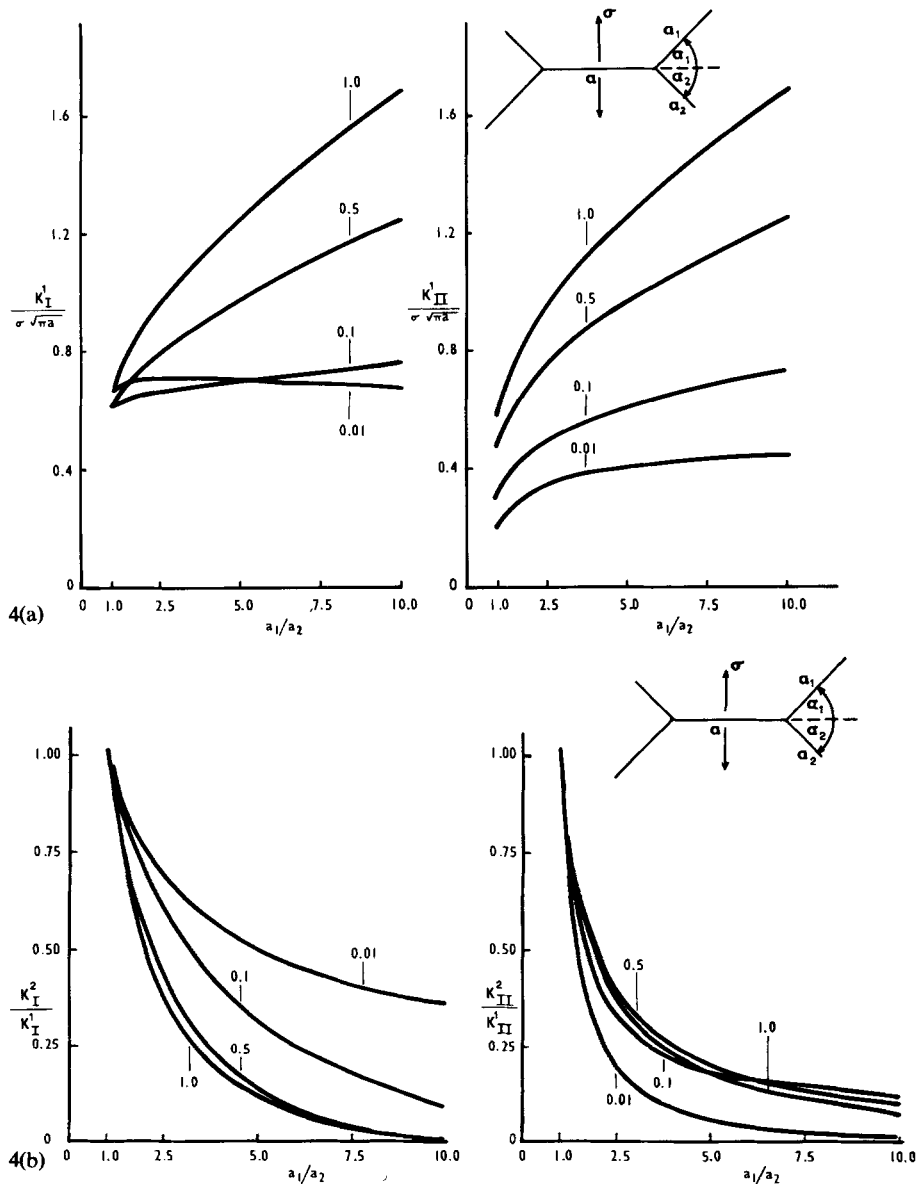


Figure 4. Stress intensity factors for asymmetrically branched crack, Fig. 2c; branching angle $\alpha_1 = 45^\circ$; (a) K_I^1 and K_{II}^1 of the longer branch vs. the ratio a_1/a_2 of the lengths of the branches for different values of a_2/a_1 ; (b) Ratios of the stress intensity factors of shorter and longer branches, K_{II}^2/K_I^1 and K_I^2/K_I^1 , vs. the ratio of their lengths a_1/a_2 for different values of a_2/a_1 .

different for the two branches. For the case of $\alpha_1 = \alpha_2 = 45^\circ$ the dependence of K_I^1 and K_{II}^1 on the ratio a_1/a_2 is shown in Fig. 4a for several different lengths of the branch 2. In Fig. 4b the ratios K_I^2/K_I^1 and K_{II}^2/K_{II}^1 are plotted as functions of a_1/a_2 for different lengths a_2 .

Stress intensity factors K_I^1 and K_{II}^1 usually increase with increasing branch length a_1 and for $a_1 \gg a_2$ they approach those of the bent crack discussed in Section 3.4. For very small a_2 a slight decrease of K_I^1 with the branch length may, however, occur as seen in Fig. 4a ($a_2/a = 0.01$, $a_1/a_2 > 1.5$). The increase takes place in this case only when $a_1 > 10a_2$. On the other hand the stress intensity factors K_I^2 and K_{II}^2 decrease rapidly with increasing length of the longer branch 1 (Fig. 4b). This occurs because the loading of branch 2 is decreased due to zero tractions along the longer branch 1. This agrees with photoelastic measurements of Theocaris [21] which show, although for a somewhat different geometry, that K_I of the short branch rapidly decreases as the length of the other branch increases. Hence, if propagation of one of the branches of a branching crack is slowed down, for example by some local obstacle, its propagation may be completely suppressed since the stress concentration at the tip of this branch will decrease rapidly due to the screening by the other branch.

3.4. Bent crack under mode I loading (Fig. 2d)

Bent crack can be considered as a special case of the branched one where branches 2 and 4 do not exist. The stress intensity factors K_I and K_{II} of branch 1 are plotted in Fig. 5 as functions of α_1 for different lengths a_1 . A similar problem was also studied in [4] assuming a branch on one side of the crack only. Comparison of this calculation with ours can be made for small a_1 as in this case the interaction between branches 1 and 3 is small. This is shown in Table 2 and very good agreement is apparent in particular for $a_1/a = 0.2$. (In Table 2 of [4] K_{II} is negative since in this calculation the bending branch was the branch 4 rather than 1).

For the case of very small branches the following analytical expressions for the stress intensity factors have been given in [8].

$$K_I = \sqrt{\pi a} F(\alpha_1) \sigma \cos \alpha_1, \quad K_{II} = \frac{1}{2} \sqrt{\pi a} F(\alpha_1) \sigma \sin \alpha_1$$

where

$$F(\alpha_1) = 3/(4 + \cos^2 \alpha_1) \left(\frac{1 - \alpha_1/\pi}{1 + \alpha_1/\pi} \right)^{\alpha_1/2\pi} \quad (5)$$

A comparison of our results for $a_1/a = 0.005$ with formulae (5) is shown in Table 3. A satisfactory agreement is obtained for $\alpha_1 < 50^\circ$ but for α_1 close to 90° the agreement is very poor. In the limit of $a_1 \rightarrow 0$ the following approximation which neglects the interaction of the branch with the original crack can be made. The normal and shear stresses at the tip of the branch may be approximated by the stresses σ'_{22} and σ'_{12} of the original crack calculated in the coordinates $x_1 y_1$ of Fig. 1. Hence when $a_1 \ll a$ we can calculate the stress intensity factors approximately as

$$K_I = \sigma'_{22} \sqrt{2\pi a_1} \quad \text{and} \quad K_{II} = \sigma'_{12} \sqrt{2\pi a_1}$$

The results of this approximation are also listed in Table 3. Although this is a rather crude approximation a reasonable agreement with our results is obtained for all angles α_1 . We can conclude, therefore, that formulae (5) derived in [8] are not generally correct but represent a good approximation for $\alpha_1 < 50^\circ$.

It is seen from Fig. 5 that unlike in the case of symmetrically branched crack (§3.1) K_{II} never changes sign and K_I possesses a maximum only for $\alpha_1 = 0$; similarly,

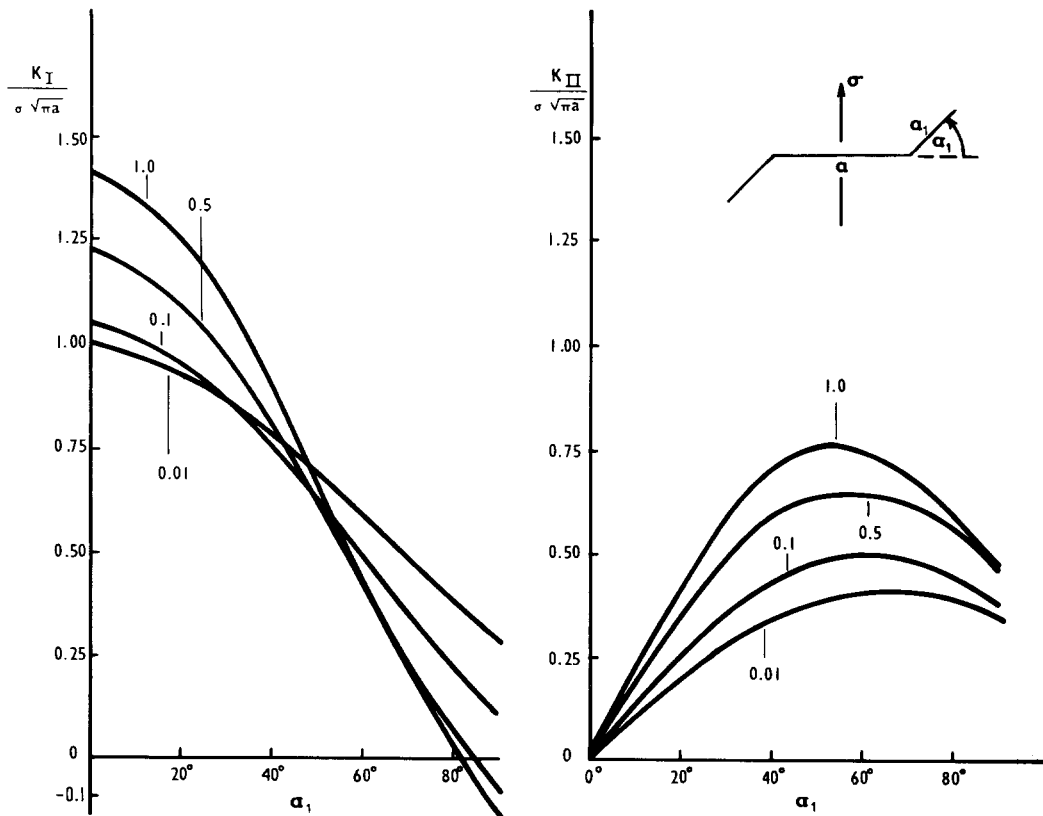


Figure 5. Stress intensity factors K_I and K_{II} of a bent crack loaded in mode I, Fig. 2d, vs. the angle of the bending branch, α_1 , for different values of a_1/a .

TABLE 3

Comparison of the stress intensity factors of the bent crack with small bending branch with formulae given in [8] and with the approximation using the stress field of the original crack (K'_I and K'_{II} have the same meaning as in Table 1).

α_1	Present work		Reference [8]		Approximation by the stress of the original crack	
	K'_I	K'_{II}	K'_I	K'_{II}	K'_I	K'_{II}
0°	1.000	0.000	1.000	0.000	1.000	0.000
10°	0.968	0.095	0.989	0.087	0.988	0.056
20°	0.930	0.184	0.930	0.173	0.955	0.168
30°	0.869	0.262	0.866	0.250	0.901	0.241
40°	0.790	0.323	0.812	0.341	0.830	0.302
50°	0.698	0.366	0.696	0.415	0.744	0.347
60°	0.598	0.388	0.548	0.474	0.650	0.375
70°	0.495	0.390	0.374	0.514	0.550	0.385
80°	0.396	0.374	0.185	0.525	0.450	0.377
90°	0.303	0.341	0.000	0.506	0.353	0.353

$(K_I)^2 + (K_{II})^2$ reaches a maximum only for $\alpha_1 = 0$. Hence, using both the criterion of the maximum tensile stress and of the maximum energy release rate the crack should never deviate from the direction perpendicular to the tensile axis and bending can, therefore, take place only due to effects of specimen boundaries, different modes of loading and/or for metallurgical reasons.

3.5. Crack branch emanating from an elliptical crack (Fig. 2e)

As in the previous case only branches 1 and 3 exist. The stress intensity factors K_I and K_{II} of the branch 1 are plotted as functions of α_1 for different root radii, $\rho = b^2/a$, of the elliptical crack, in Fig. 6a for $a_1 = 0.01a$ and in Fig. 6b for $a_1 = 0.5a$ ($\rho = 0$ corresponds to the sharp original crack). These results provide an approximation to the case of one crack branch emanating from a notch.

It is seen that for $a_1 \ll a$ both K_I and K_{II} are substantially smaller than for the case of the sharp crack whilst for a_1 comparable with a the difference is diminishing. As in the previous case both K_I and $(K_I)^2 + (K_{II})^2$ possess only one maximum at $\alpha_1 = 0$ and thus within the framework of elasticity theory a sharp crack will always propagate parallel to the principal axis of the ellipse. However, if, for example, crack deviation occurs due to metallurgical reasons it will be easier when the original crack is blunt rather than sharp, since as seen in Fig. 6a, the larger the root radius the more slowly K_I decreases as α_1 increases.

3.6. Bent crack under mixed mode loading (Fig. 2f)

The tensile axis is inclined by an angle $\alpha < 90^\circ$ to the plane of the original crack and only branches 2 and 4 exist. For the case of $\alpha = 45^\circ$ the stress intensity factors K_I and K_{II} of branch 2 are plotted as functions of α_2 in Fig. 7. The factor $(K_I)^2 + (K_{II})^2$ determining the energy release rate is plotted in Fig. 8. It is seen that K_{II} always decreases and eventually changes sign with increasing angle α_2 and K_I reaches a maximum when $K_{II} = 0$. However, the angle for which $(K_I)^2 + (K_{II})^2$ reaches a maximum does not generally coincide with that for which K_I is a maximum. Fig. 9 shows the dependence on the branch length of the angle $\theta = \alpha + \alpha_2$ between the branch and the tensile axis, for which $(K_I)^2 + (K_{II})^2$ reaches a maximum (full curve), and also of that for which K_I is a maximum (dashed curve).

The two criteria used in the study of the crack propagation under mixed mode loading assume that the propagation takes place either in the direction of the maximum tensile stress (i.e. maximum K_I) or in the direction of the maximum energy release rate. It is seen from Fig. 9 that these two criteria are in general different although they will give a similar answer when the bent branch is small. This is in variance with the results of Nuismer [22] who concludes that in the limit of $a_2 \rightarrow 0$ these criteria are identical. In his calculations it was assumed that in this limit the stress field at the tip of the branch is the same as that of the original crack. This limiting process is not, however, exact and applies only approximately as seen from Table 3 where a similar assumption was made when calculating the third column.

Using any of the two above mentioned criteria the crack loaded in tension with the tensile axis inclined by 45° to the crack plane will first tend to propagate in a direction approximately 12° from the direction perpendicular to the tensile axis. As the crack length increases this angle will decrease and the energy release rate criterion predicts that when $a_2 \approx a$ the propagation will be practically in the direction per-

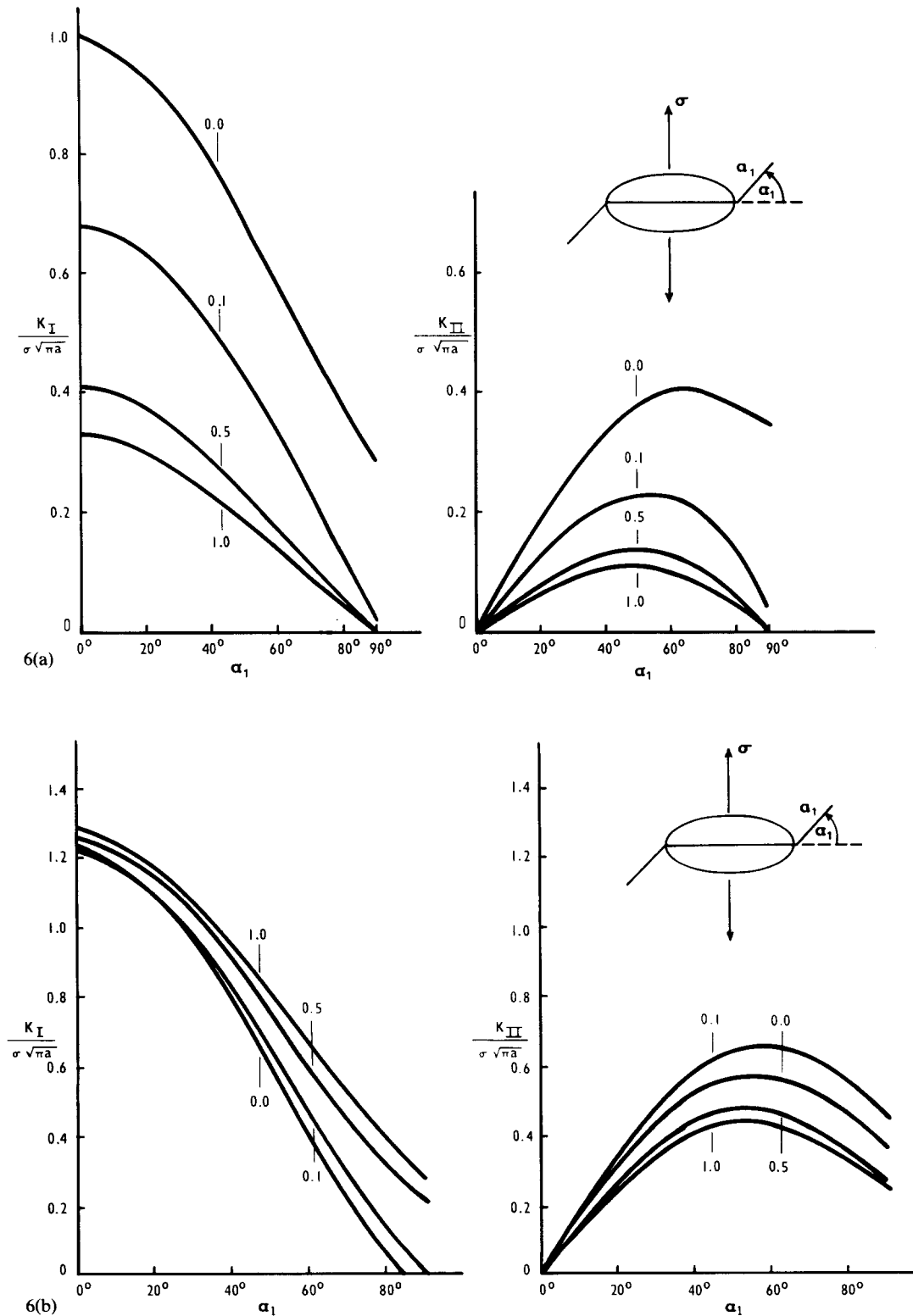


Figure 6. Stress intensity factors K_I and K_{II} of a crack branch emanating from an elliptical crack, Fig. 2e, vs. the angle of branch inclination, α_1 for different root radii, ρ/a , at the tip of the crack; (a) Crack branch length $a_1 = 0.01a$; (b) Crack branch length $a_1 = 0.5a$.

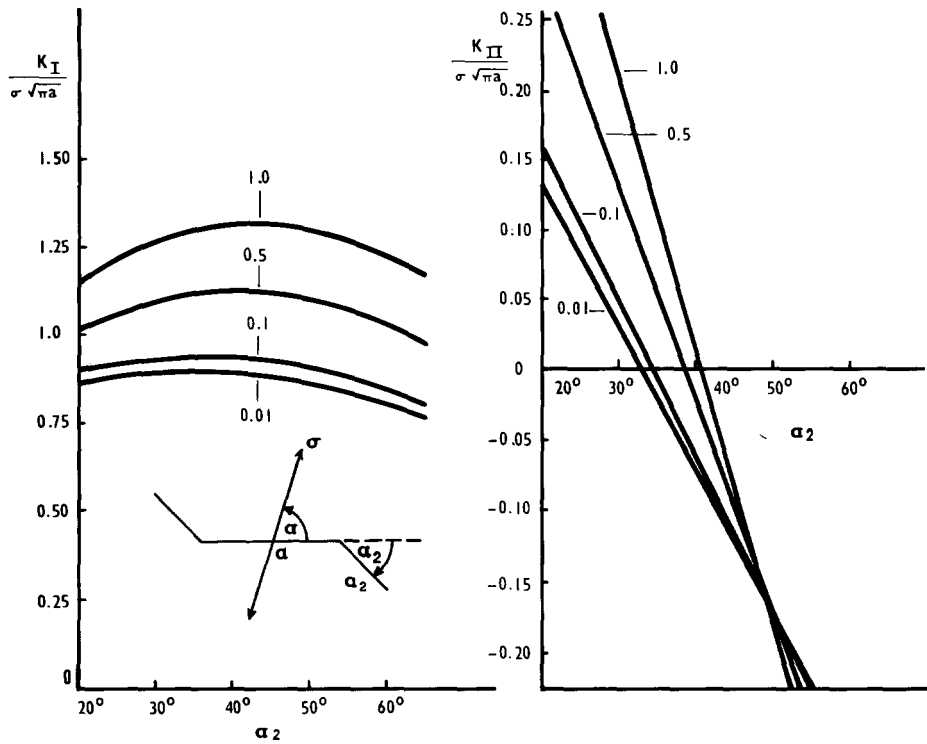


Figure 7. Stress intensity factors K_I and K_{II} of a bent crack loaded in tension with the tensile axis inclined by 45° to the plane of the original crack, Fig. 2f, vs. the angle α_2 .

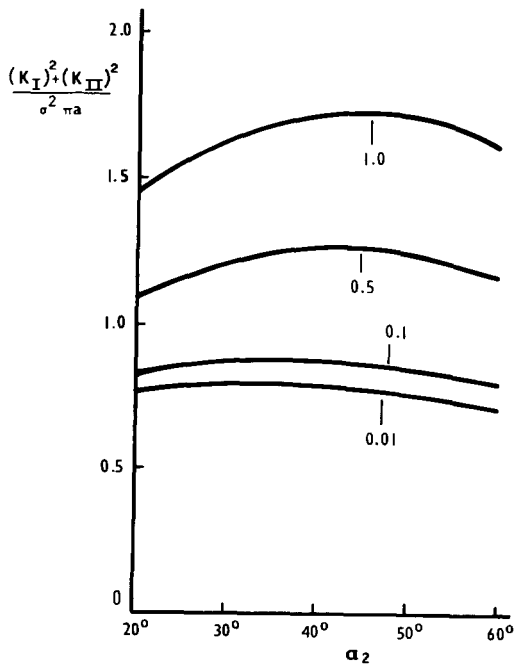


Figure 8. $(K_I)^2 + (K_{II})^2$ for the case of Fig. 2f ($\alpha = 45^\circ$) vs. the angle, α_2 .

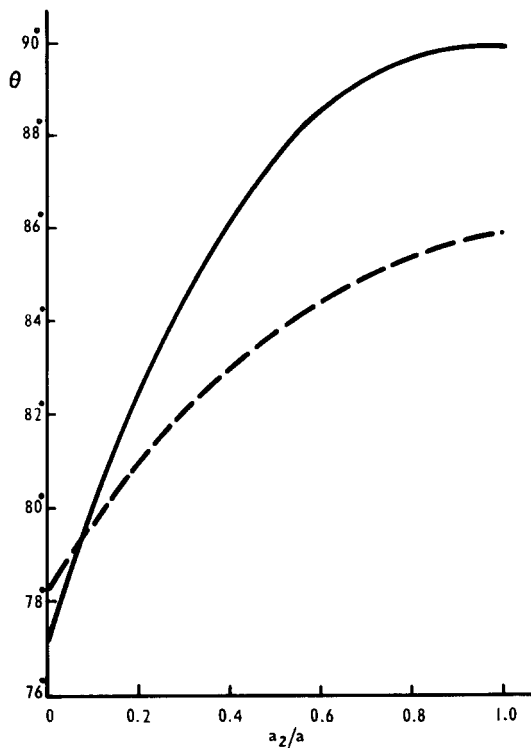


Figure 9. The angle $\theta = \alpha_2 + 45^\circ$ between the bending branch and the tensile axis, Fig. 2f, vs. the length of the bending branch, a_2 , for (i) maximum energy release rate (full line), (ii) maximum tensile stress at the tip of the branch (dashed line).

pendicular to the tensile axis whilst the other criterion predicts for this case a deviation of about 4° .

4. Conclusions

1. A numerical method, which employs a dislocation model of cracks, has been developed to calculate the stress intensity factors of branched and bent cracks and crack branches emanating from elliptical cracks. The precision of the method is estimated to be $\pm 2\%$.
2. Branching of the crack loaded in mode I always leads to substantial decreases in both K_I and the energy release rate and may, therefore, be responsible for arresting propagating cracks. In the case of the symmetrically branched crack K_I reaches a local maximum for inclination angles between 8° and 24° depending on the branch length. Assuming that the branches tend to propagate in the direction of the maximum tensile stress they may first "repel" each other if the starting branching angle is smaller than 24° but "attraction" will take place as the length of the branches increases. The same applies for the symmetrical crack branches emanating from an elliptical crack.
3. If the branches are of different length the shorter one will be screened by the longer one so that the stress intensity factors of the shorter branch will be substantially decreased. Hence, if this situation occurs during branching, propagation of the shorter branch may be completely suppressed by this screening effect.
4. Bending of a crack loaded in mode I may occur only due to the influence of

specimen boundaries or for metallurgical reasons. The same applies for cracks emanating from elliptical cracks. Deviation of a crack with large root radius will be easier than for a sharp crack.

5. When a crack is inclined to the direction of the tensile axis it will not propagate immediately in the direction perpendicular to the tensile axis but in a direction which makes an angle smaller than 90° with it and will tend to the perpendicular direction as the crack length increases. The criteria of the maximum tensile stress and maximum energy release rate used in the study of the direction of crack propagation under mixed mode loading are generally different and give similar results only when the bent segment of the crack is small.

Appendix I

In order to calculate stresses needed in Eqn. (2) we employ the theory of complex potentials for plane strain [15]. The stress tensor σ_{ij} calculated in the coordinates x, y (Fig. 1) is then expressed in terms of two complex functions $\phi(z)$ and $\psi(z)$ of the variable $z = x + iy$:

$$\sigma_{11} + \sigma_{22} = 2(\phi'(z) + \overline{\phi'(z)}) \quad (\text{AI.1})$$

$$\sigma_{22} - \sigma_{11} + 2i\sigma_{12} = 2(\bar{z}\phi''(z) + \psi'(z))$$

where the bar means complex conjugate. The corresponding stresses τ_{mn} are obtained by tensor transformation into coordinates x_k, y_k . To calculate the crack and image stresses of the dislocations we use the conformal mapping

$$z = R\left(\zeta + \frac{m}{\zeta}\right) \quad (\text{AI.2})$$

where $R = (a + b)/2$ and $m = (a - b)/(a + b)$ which maps the ellipse of axes a, b in the plane z onto unit circle in the plane ζ . The complex potentials which determine the stress field of the elliptical crack loaded in tension with the tensile axis inclined at an angle α to the major axis of the ellipse were given in [15].

$$\begin{aligned} \phi^{c+\text{appl}}(\zeta) &= \frac{\sigma R}{4} \left(\zeta + \frac{2e^{2i\alpha} - m}{\zeta} \right) \\ \psi^{c+\text{appl}}(\zeta) &= \frac{\sigma R}{4} \left(\zeta e^{-2i\alpha} + \frac{e^{2i\alpha}}{\zeta} - \frac{(1 + m^2)(e^{2i\alpha} - m)}{m} \frac{\zeta}{\zeta^2 - m} \right) \end{aligned} \quad (\text{AI.3})$$

The potentials which determine the image stress field were derived in [23]. When the dislocation with the Burgers vector $\mathbf{b} = (b_x, b_y, 0)$ lies parallel to z and intersects the xy plane in the point x_d, y_d , they are

$$\begin{aligned} \phi^{\text{dim}}(\zeta) &= 2\gamma \ln \zeta - \gamma \ln(\zeta - m/\zeta_d) - \gamma \ln(\zeta - 1/\bar{\zeta}_d) \\ &\quad + \frac{\bar{\gamma}}{\zeta_d \bar{\zeta}_d (\zeta_d^2 - m)} [\zeta_d(1 + m\bar{\zeta}_d^2) - \bar{\zeta}_d(\zeta_d^2 + m)] \frac{1}{\zeta - 1/\bar{\zeta}_d} \\ \psi^{\text{dim}}(\zeta) &= 2\bar{\gamma} \ln \zeta - \bar{\gamma} \ln(\zeta - m/\zeta_d) - \bar{\gamma} \ln(\zeta - 1/\bar{\zeta}_d) \\ &\quad + \frac{\gamma}{\zeta_d \bar{\zeta}_d (\zeta_d^2 - m)} [\bar{\zeta}_d(\zeta_d^2 + m^3) - m\zeta_d(\bar{\zeta}_d^2 + m)] \frac{1}{\zeta - m/\zeta_d} \\ &\quad - \zeta \frac{1 + m\zeta^2}{\zeta^2 - m} [\phi^{\text{dim}}(\zeta)]' \end{aligned} \quad (\text{AI.4})$$

where ζ_d is the conformal transform of $z_d = x_d + iy_d$ using Eqn. (AI.2) and $\gamma = [G/4\pi(1-\nu)](b_y - ib_x)$.

Appendix II

When calculating $\tau_{mn}^{dim,i}$ in Eqn. (3) we regarded the unit dislocations as singular, positioned in the centre of corresponding intervals. The evaluation of these stresses is then straightforward when using Eqns. (AI.4). On the other hand when calculating $\tau_{mn}^{d,i}$ we assumed constant dislocation density $\rho_j^l = 1/\Delta x_j$ in each interval l of the branch j . However, when calculating the stress at the points of branch j due to the dislocation in the interval n of this branch we took a continuous dislocation distribution density of the form consistent with Eqn. (1), namely

$$\rho_j^n = \frac{1}{2\sqrt{\Delta x_j}} \frac{1}{\sqrt{a_j - x_j}} \quad (\text{AII.1})$$

This approximation to the density near the branched crack tips was found to be essential in order to obtain sufficient precision.

If the stress field of a singular dislocation positioned at t_j is $\sigma_{mn}^{ds}(x_j - t_j, y_j)$ than the stress field of the continuous distribution of the dislocations of density

$$\rho_j(t_j) \text{ is } \sigma_{mn}^d(x_j, y_j) = \int \rho_j(t_j) \sigma_{mn}^{ds}(x_j - t_j, y_j) dt_j$$

where the integration extends over the region of the dislocation distribution. Using the standard formulae for the stress field of singular dislocations (e.g. [24]) the stresses $\sigma_{mn}^{d,i}$ have been evaluated using the above approximation for the dislocation densities. In the coordinate system x_j, y_j the stress field due to the unit dislocation in the interval $l < n$ of branch j is for $y_j \neq 0$.

$$\begin{aligned} \sigma_{11}^{d,1} &= -B \left[\frac{1}{2} \ln \left(\frac{1+z_1^2}{1+z_2^2} \right) + \frac{3-z_1^2}{4(1+z_1^2)} - \frac{3-z_2^2}{4(1+z_2^2)} \right] \\ \sigma_{22}^{d,1} &= -B \left[\frac{1}{2} \ln \left(\frac{1+z_1^2}{1+z_2^2} \right) + \frac{3z_1^2-1}{4(1+z_1^2)} - \frac{3z_2^2-1}{4(1+z_2^2)} \right] \\ \sigma_{12}^{d,1} &= B \left[\frac{z_1}{1+z_1^2} - \frac{z_2}{1+z_2^2} \right] \end{aligned} \quad (\text{AII.2a})$$

when the Burgers vector is perpendicular to the branch and

$$\begin{aligned} \sigma_{11}^{d,2} &= -B \left[2(\arctan z_1 - \arctan z_2) - \frac{z_1}{1+z_1^2} + \frac{z_2}{1+z_2^2} \right] \\ \sigma_{22}^{d,2} &= -\sigma_{12}^{d,1} \\ \sigma_{12}^{d,2} &= -\sigma_{22}^{d,1} \end{aligned} \quad (\text{AII.2b})$$

when the Burgers vector is parallel to the branch.

$$B = [G/2\pi(1-\nu)] \frac{1}{\Delta x_j}, \quad z_1 = (x_j - x_j^l + \Delta x_j/2)/y_j$$

and $z_2 = (x_j - x_j^l - \Delta x_j/2)/y_j$,
when $y_j = 0$ and $l < n$.

$$\sigma_{22}^{d,1} = \sigma_{12}^{d,2} = B \ln \left| \frac{x_j - x_j^l - \Delta x_j/2}{x_j - x_j^l + \frac{\Delta x_j}{2}} \right| \quad (\text{AII.2c})$$

$$\sigma_{22}^{d,2} = \sigma_{12}^{d,1} = 0$$

and for $l = n$

$$\sigma_{22}^{d,1} = \sigma_{12}^{d,2} = -\frac{B\sqrt{(\Delta x_j)}}{2\sqrt{x_j - x_j^l + \Delta x_j/2}} \ln \left| \frac{\sqrt{x_j - x_j^l + \Delta x_j/2} + \sqrt{\Delta x_j}}{\sqrt{x_j - x_j^l + \Delta x_j/2} - \sqrt{\Delta x_j}} \right| \quad (\text{AII.2d})$$

$$\sigma_{22}^{d,2} = \sigma_{12}^{d,1} = 0$$

Using Eqns. (AII.2a–d) and (AI.4) the stress tensors are calculated in the coordinate systems x_j, y_j and x, y , respectively. The stress tensor $\tau_{mn}^{d,i}$ needed in any other system x_i, y_i is obtained by obvious tensor transformation.

The system of linear Eqns. (3a, b) can be solved by standard numerical methods; the number of intervals $n = 30$ was always found to be sufficient. However, when solving these equations oscillations of a numerical character occurred similarly to those mentioned in [23] so that $b_{i,1}^i < b_{i,2}^i > b_{i,3}^i < b_{i,4}^i > b_{i,5}^i \dots$. These oscillations, although small, are not negligible and non-oscillatory solution was obtained as follows: dislocations 1, 2, ..., $n-2$ were replaced by $(n-2)/2$ dislocations of Burgers vectors $b_{i,1'}^i = b_{i,1}^i + b_{i,2}^i$, $b_{i,2'}^i = b_{i,3}^i + b_{i,4}^i \dots$ which are contained in corresponding intervals of length $2\Delta x_j$. The following four equations were then solved for the Burgers vectors $b_{i,n'}^i$ where $n' = n/2$, of the dislocations at the tip of the branches:

$$\begin{aligned} & \sum_{j=1}^2 \sum_{i=1}^2 b_{i,n'}^i [\tau_{mn}^{d,i}(x_k^{n'}, x_j^{n'}) - \tau_{mn}^{d,i}(x_k^{n'}, x_{j+2}^{n'}) \\ & + \tau_{mn}^{\text{dim},i}(x_k^{n'}, x_j^{n'}) - \tau_{mn}^{\text{dim},i}(x_k^{n'}, x_{j+2}^{n'})] \\ & = \sum_{l'=1}^{n'-1} \sum_{j=1}^2 \sum_{i=1}^2 b_{i,l'}^i [\tau_{mn}^{d,i}(x_k^{n'}, x_{j+2}^{l'}) - \tau_{mn}^{d,i}(x_k^{n'}, x_j^{l'}) \\ & + \tau_{mn}^{\text{dim},i}(x_k^{n'}, x_{j+2}^{l'}) - \tau_{mn}^{\text{dim},i}(x_k^{n'}, x_j^{l'})] - \tau_{mn}^{c+\text{appl}}(x_k^{n'}) \end{aligned} \quad (\text{AII.3})$$

mn is 22 and 12, respectively, and $k = 1, 2$ numbers the branches, $b_{i,n'}^i$ was used in Eqn. (4) with Δx_j replaced by $2\Delta x_j$. This procedure leads in all testing cases to an improvement of the precision, from $\pm 6\%$ using $b_{i,n}^i$ (obtained from the solution of Eqns. (3a, b)) to $\pm 2\%$.

Acknowledgements

The author would like to thank Dr. G.G. Ghell for many helpful discussions and for his comments on the manuscript. This work was performed at Central Electricity Research Laboratories and is published by permission of the Central Electricity Generating Board.

REFERENCES

- [1] F. Erdogan, *Fracture*, Vol. II, ed. H. Liebowitz, Academic Press (1968) 498.
- [2] M.O. Speidel, *Theory of Stress Corrosion Cracking*, NATO Conference Ericeira (1971) 289.
- [3] C.S. Carter, *Engineering Fracture Mechanics*, 3 (1972) 1.
- [4] H. Kitagawa, R. Yuuki and T. Ohira, *Engineering Fracture Mechanics*, 7 (1975) 515.
- [5] I.M. Austen, R. Brook and J.M. West, *International Journal of Fracture*, 12 (1976) 253.
- [6] F. Erdogan and G.C. Sih, *Transactions of ASME, Series D., Journal of Basic Engineering*, 85 (1963) 519.

- [7] J.G. Williams and P.D. Ewing, *International Journal of Fracture Mechanics*, 8 (1972) 441.
- [8] M.A. Hussain, S.L. Pu and J. Underwood, *Assoc. National Symposium on Fracture Mechanics II*, ASTM Special Technical Publication 560 (1973) 2.
- [9] G.C. Sih, *Transaction of ASME, Series E., Journal of Applied Mechanics*, 24 (1965) 51.
- [10] E. Smith, *Journal of Mechanics and Physics of Solids*, 16 (1968) 329.
- [11] T. Nahayama, *Transactions JSME*, 39 (1973) 322.
- [12] J.R. Willis, *International Journal of Fracture*, 11 (1975) 489.
- [13] H. Andersson, *Journal of Mechanics and Physics of Solids*, 17 (1969) 405.
- [14] G.C. Sih and P.C. Paris, *Transactions of ASME, Series E*, 306 (1962).
- [15] N.I. Muskhelishvili, *Some Basic Problems of the Mathematical Theory of Elasticity*, Gos. Izdat, Fismatgiz, Moscow (1966) (English translation by J.R.M. Radok, Noordhoff, Leyden, 1975).
- [16] H. Andersson, *Journal of Mechanics and Physics of Solids*, 18 (1970) 437.
- [17] H. Kitagawa and R. Yuuki, *Transactions JSME*, 41 (1975) 346.
- [18] S.N. Chatterjee, *International Journal of Solids and Structures*, 11 (1975) 521.
- [19] E. Smith, *International Journal of Fracture Mechanics*, 1 (1965) 204.
- [20] J.F. Kalthoff, *International Journal of Fracture Mechanics*, 7 (1971) 478.
- [21] P.S. Theocaris, *Journal of Mechanics and Physics of Solids*, 20 (1972) 265.
- [22] R.J. Nuismer, *International Journal of Fracture*, 11 (1975) 245.
- [23] V. Vitek, *Journal of Mechanics and Physics of Solids*, 24 (1976) 67.
- [24] J.P. Hirth and J. Lothe, *Theory of Dislocations*, McGraw-Hill, New York-London (1968).